Gold Price Forecasting

- Applied Time Series Analysis | B.Tech AI Semester VII | 2026-27
- ARIMA / SARIMA analysis with an INR-per-gram application
- 18K, 20K, 22K and 24K fine-gold estimates
- Priyanshu Modak (I080) | Vishesh Dokania (I019) | Parth Zanwar (I074)

The questions we investigated

- Can past prices support useful daily forecasts?
- Does ARIMA or weekly SARIMA improve on tomorrow = today?
- How does uncertainty change with the forecast horizon?
- How can forecasts be displayed for Indian users?

All 11 required ATSA steps

- 1 Collection | 2 EDA | 3 Missing values
- 4 ADF/stationarity | 5 Transformations | 6 Seasonality
- 7 ACF/PACF | 8 Model family | 9 Lags
- 10 AIC/BIC | 11 Streamlit deployment

Real data and provenance

- Yahoo Finance GC=F: rolling COMEX gold futures
- 2,971 source rows | 02 Jan 2015 to 26 Oct 2026
- Native price: USD per troy ounce
- USD-INR source: INR=X; dated snapshot retained

Architecture: analysis and application

- Notebook: collect -> clean -> analyse -> select -> evaluate
- Artifacts: charts, CSVs, fitted model and metadata
- App: fetch/cache -> refit ARIMA -> forecast -> convert
- GitHub holds source; Streamlit provides hosting

Cleaning and chronological split

- Sort/deduplicate; require positive numeric closes
- Forward-fill 111 missing weekdays: 3.60%
- Regular series: 3,082 business weekdays
- Train 2,982 through 08 Jun | Test 100 from 09 Jun 2026

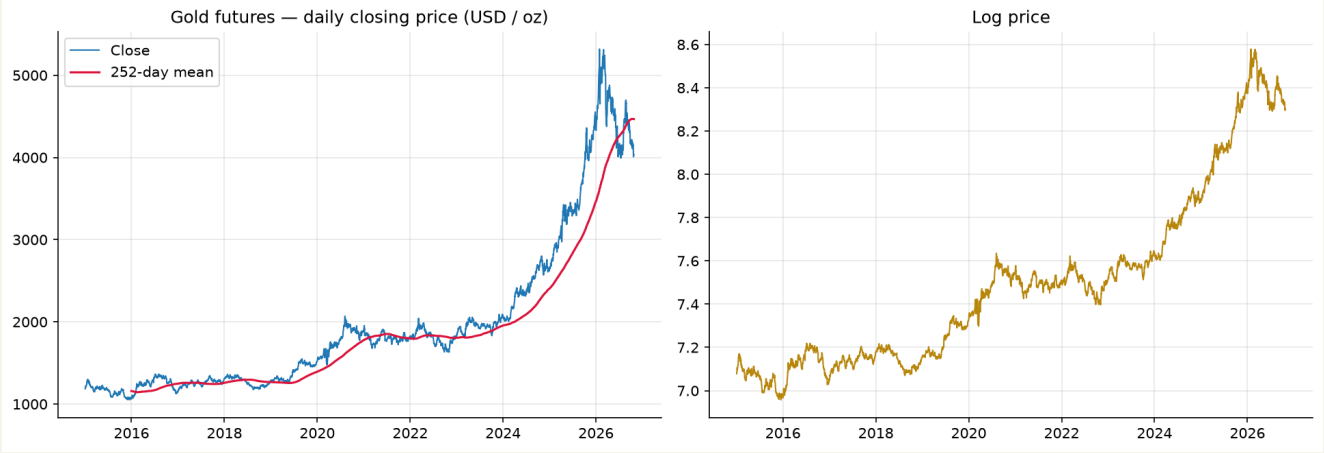
EDA: price levels and dependence

- Source mean: USD 1,966.46 per troy ounce

- Source-price standard deviation: USD 956.73

- Lagged prices move together; returns need separate study

- A rising price level alone does not establish predictability



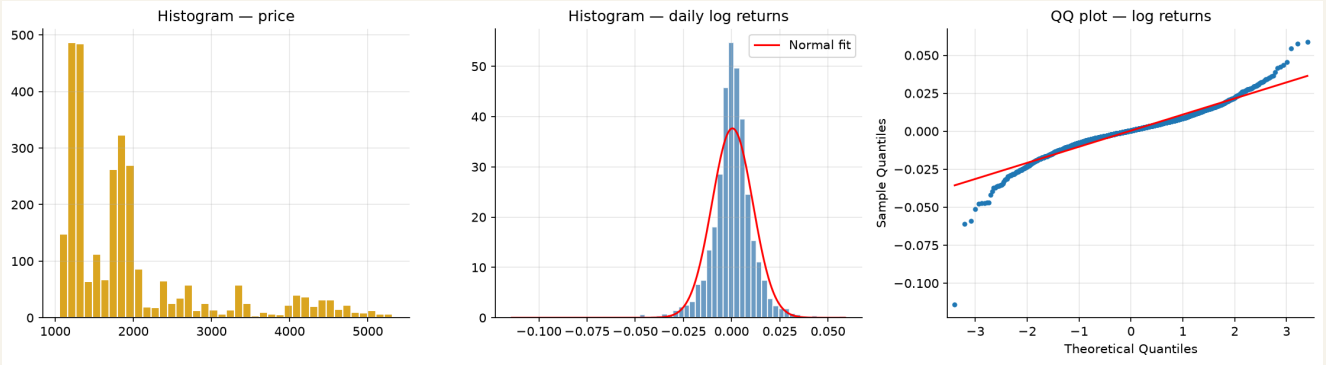
EDA: distribution and heavy tails

- Source-return excess kurtosis: 7.7843

- Jarque-Bera rejects normality; QQ tails depart

- Source-return annualised volatility: about 16.8%

- Heavy tails limit unqualified Gaussian interval claims



Stationarity: opposite null hypotheses

- Price: ADF p=.9755; KPSS lower reported bound .01
- Log price: ADF p=.9897; KPSS lower bound .01
- Log return: ADF p about 6.83e-27; KPSS upper bound .10
- The sample supports first-differenced logs: d=1

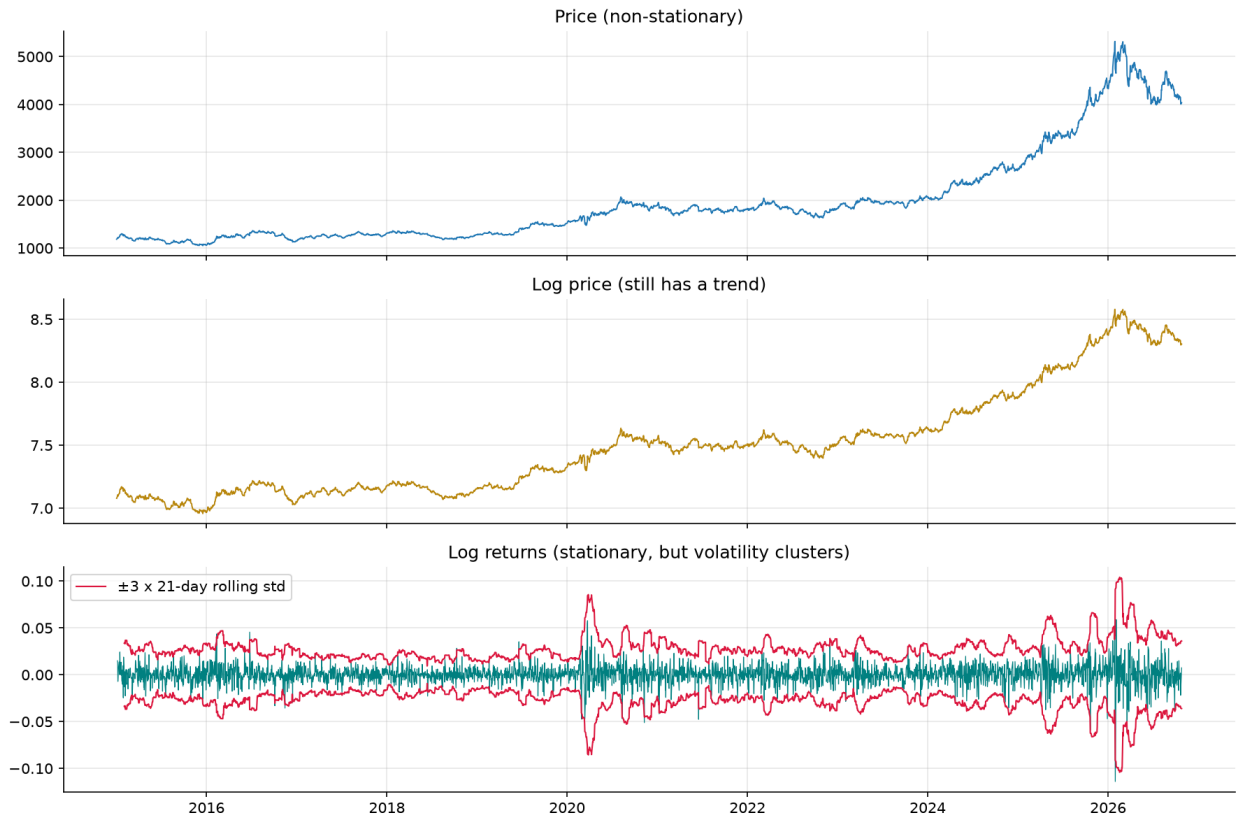
Transformation and selected equation

- z(t) = log(P(t)); r(t) = z(t) - z(t-1)

- ARIMA(p,1,q) models short-memory dynamics in r(t)

- Selected: z(t) = z(t-1) + drift + innovation

- exp(log forecast) is a conditional median



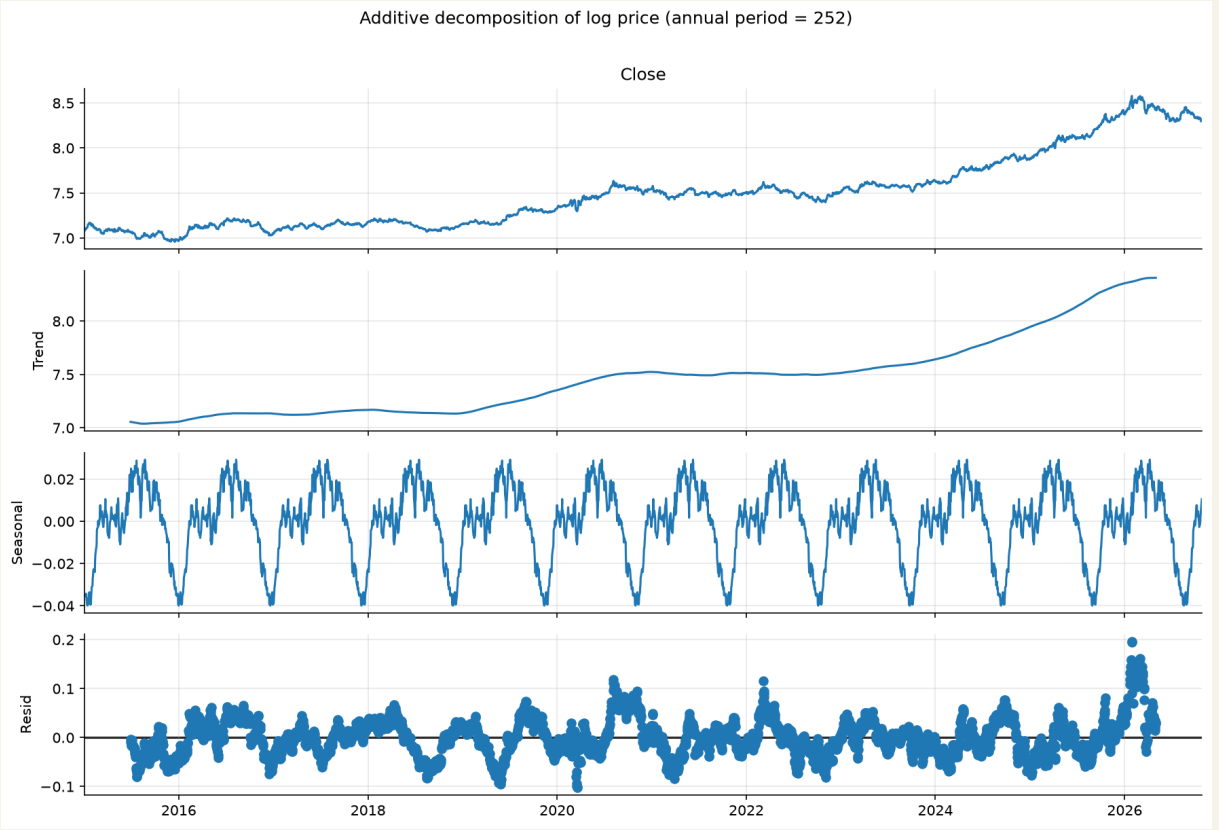
Seasonality: examine rather than assume

- Strength: 5 days .0002 | 21 days .0084

- 252-business-day strength: .1747

- Annual decomposition is descriptive evidence only

- A weekly SARIMA competitor is evaluated



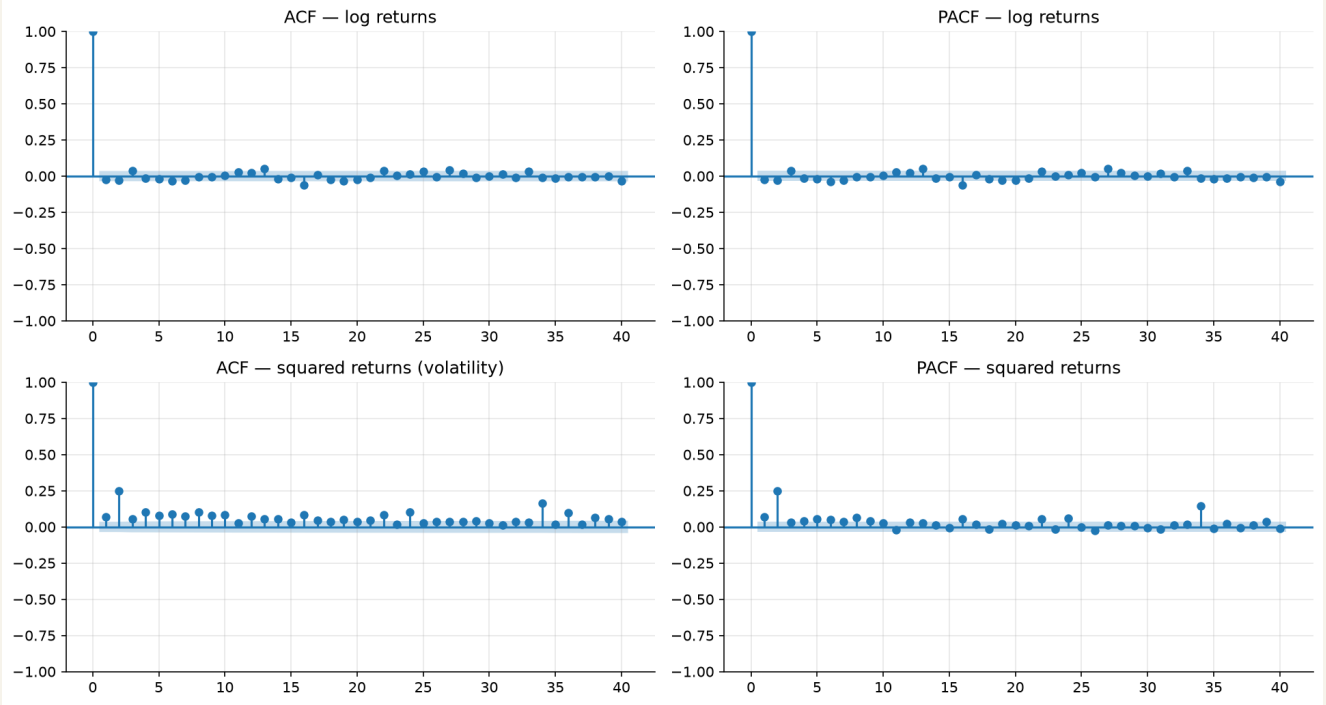
ACF/PACF and model identification

- AR: PACF cutoff; ACF usually decays

- MA: ACF cutoff; PACF usually decays

- ARMA: both decay; seasonal spikes motivate SARIMA

- Squared-return dependence motivates variance analysis



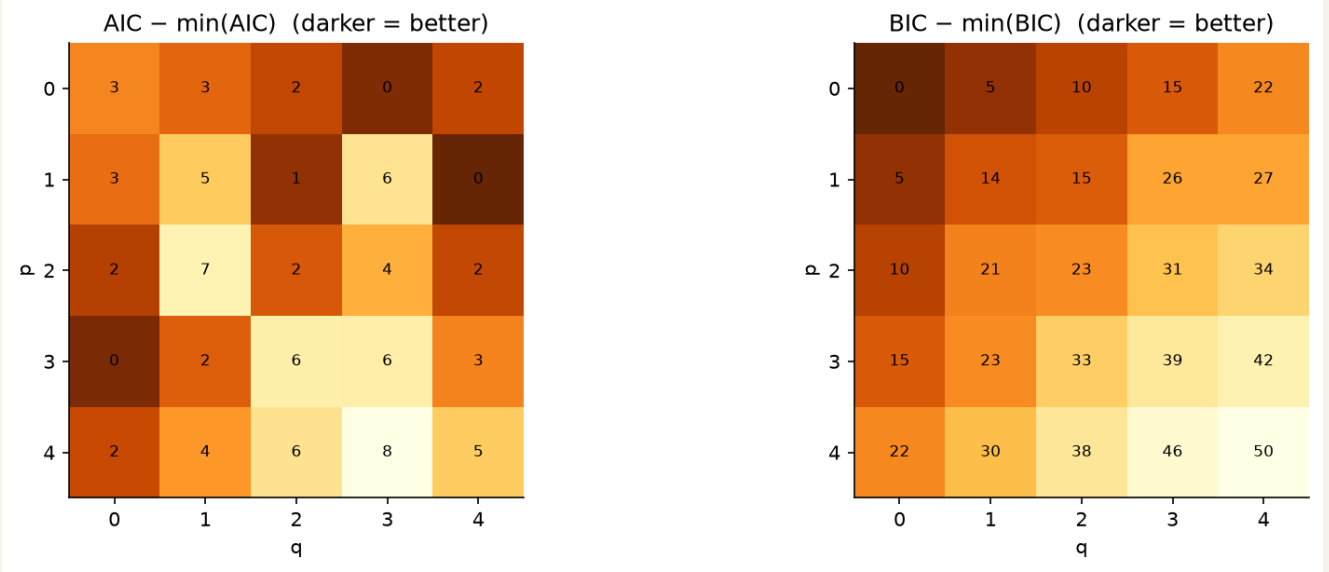
The recorded 25-model search

- p=0..4; d=1; q=0..4 -> 25 candidate orders

- Same training log-price series; drift in every candidate

- AIC selects (1,1,4); BIC selects (0,1,0)

- Convergence warnings occurred: rankings need a caveat



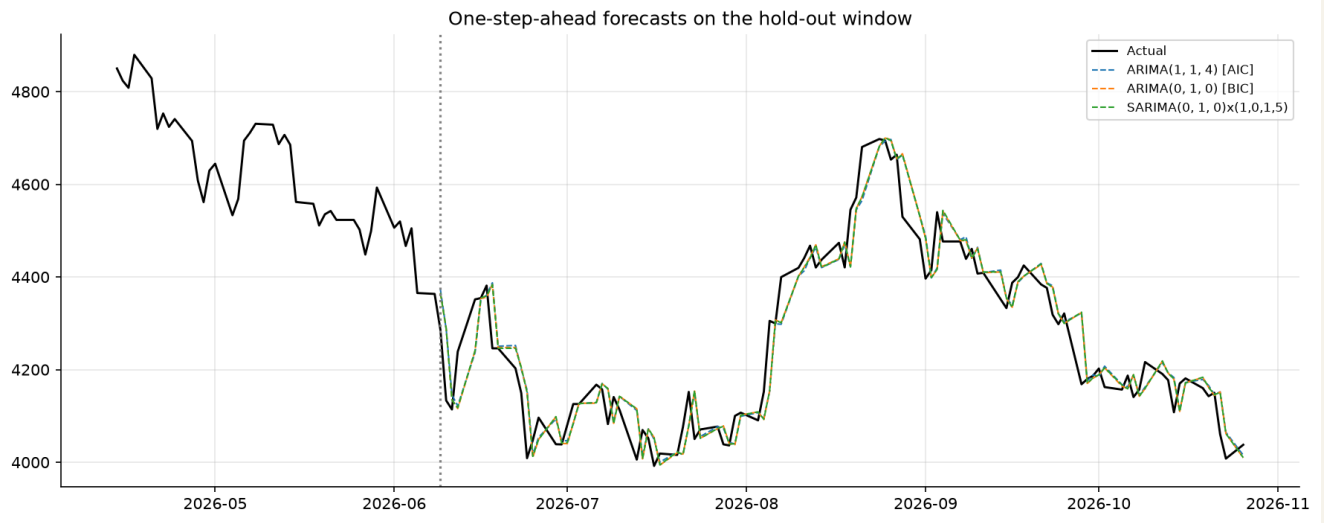
Two evaluation designs

- One-step: forecast, then observe for later predictions

- Model parameters stay fixed during hold-out updates

- Fixed-origin: forecast 100 dates with no test updates

- The app refits available history and forecasts 5..60 weekdays



Measured performance: naive wins RMSE

- Naive 60.5337 | BIC ARIMA 60.6723 USD/oz
- AIC ARIMA 60.6684 | SARIMA 60.6085 USD/oz
- BIC MAPE 1.0778%; directional agreement 49%
- HAC loss-difference p=.4639: BIC ARIMA vs naive

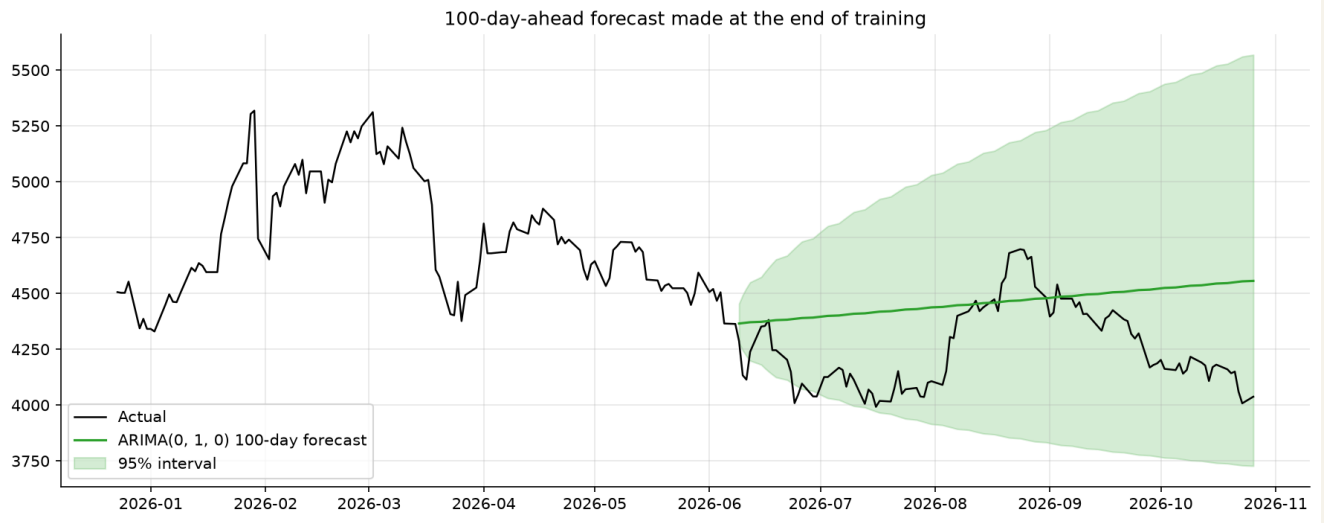
Intervals and residual diagnostics

- Fixed-origin 100-day interval coverage: 94/100

- Residual Ljung-Box p=.08397 at lag 10; .000589 at lag 20

- ARCH-LM p about 6.28e-45; normality rejected

- Longer-lag dependence and changing variance remain



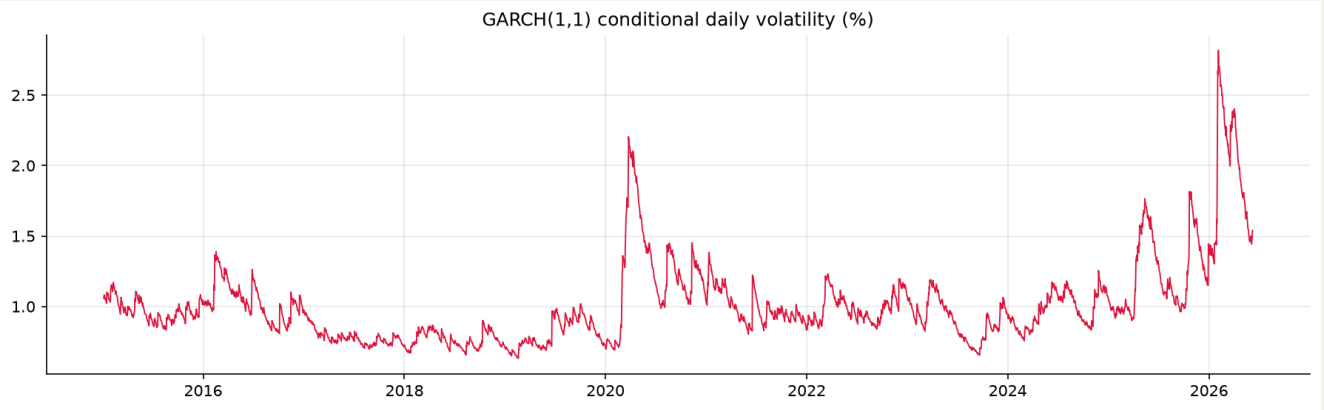
Optional GARCH volatility extension

- Variance = omega + alpha*past shock^2 + beta*past variance

- Training alpha=.0359; beta=.9583; sum=.9942

- Student-t innovations fitted to percentage log returns

- Final normal-quantile bands remain approximate



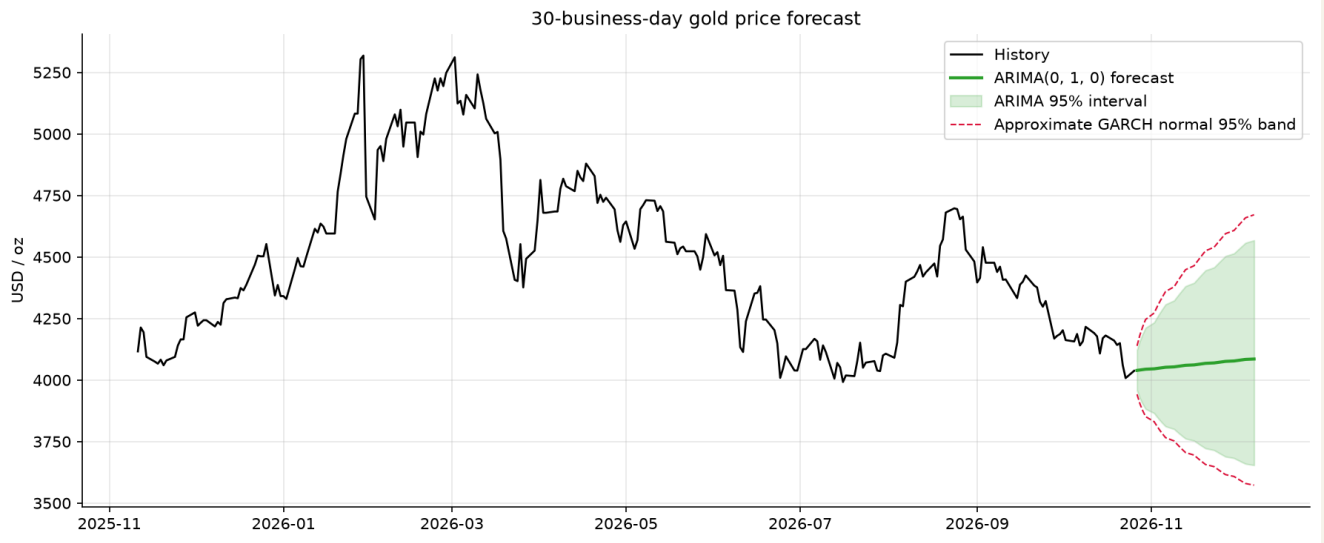
Final saved forecast

- Selected ARIMA refitted on all 3,082 observations

- 30-business-day snapshot saved as a forecast CSV

- 27 Oct 2026 median: USD 4,039.29 per troy ounce

- ARIMA interval: USD 3,957.84 to 4,122.41



INR per gram and all four karats

- INR/g = USD/oz x INR/USD / 31.1034768 x fineness
- 18K .750 | 20K .833 | 22K .916 | 24K .999
- Dated automatic FX or a positive manual override
- Same factor scales history, median and both bounds

What the Indian-price display means

- Selected FX is held fixed for all future dates
- History is revalued at that same selected rate
- Future FX risk, GST, duties and making charges excluded
- Metal-value estimates are not jewellery-shop quotations

Professor demo: deliberate actions

- Start 24K, 30 weekdays, ARIMA(0,1,0)
- Switch 22K -> 20K -> 18K; inspect comparison table
- Compare horizons 5 and 60; change FX and order
- Export CSVs; explain valid upload and real-data fallback

All finite setting combinations

- 4 karats x 13 history values x 56 horizons x 72 orders
- 72 orders = 6 p choices x 2 d choices x 6 q choices
- 209,664 discrete tuples are fully enumerated
- Enumeration is not 209,664 executed model fits

Decision combinations and evidence

- ADF rejects + KPSS does not: stationarity supported
- ADF does not reject + KPSS rejects: investigate differencing
- Both reject or neither rejects: evidence is inconclusive
- ARCH rejection: inspect variance, not just mean dynamics

Code walkthrough and deployment

- streamlit_app.py: root cloud entry point
- app.py: loaders -> ARIMA -> INR conversion -> exports
- run_analysis.py and make_report.py: analysis/report generation
- Public GitHub source; Streamlit deployment confirmed by user

Limitations and next steps

- No demonstrated RMSE advantage over naive
- Convergence flags and rolling-origin checks need improvement
- No forecast of FX, interest rates, inflation or news
- Next: aligned INR target, trading calendar, calibrated intervals

Conclusion and handover

- Complete ATSA workflow on preserved real market data
- d=1 supported; BIC favours a drifted random walk
- Measured errors and diagnostic failures reported honestly
- Working INR/g app with four purity views

Backup: order families and trend rules

- (0,1,0): random walk with log drift
- (p,1,0): AR in differences | (0,1,q): MA in differences
- (p,1,q): ARMA in first-differenced log prices
- App d=2 is available; its deterministic trend is n

Backup: failure and fallback combinations

- Gold live fails + real cache: labelled cached prices
- FX live fails + snapshot: dated saved exchange rate
- No FX source: require a positive manual rate
- Bad CSV / insufficient data: stop rather than fabricate

Backup: four questions to answer precisely

- Does 1.1% MAPE mean 98.9% accuracy?
- Why can naive outperform fitted ARIMA?
- Why does the app differ from the notebook?
- Why is a 24K result different from a jeweller quote?

References and assistance disclosure

- Yahoo Finance GC=F and INR=X; source dates disclosed
- statsmodels, arch and Forecasting: Principles and Practice
- NIST conversion and BIS fineness grades
- AI-assisted implementation; state actual team contributions